Medvale Internal Medicine - Chapter 10F

Rheumatology Review: Integrated Cases, Diagnostic Algorithms, and High-Yield Differentials

Purpose: This chapter consolidates Part X Rheumatology and Immunology into practical illness scripts. It emphasizes pattern recognition, diagnostic sequencing, emergency triage, treatment logic, and medication safety. The text is intentionally paragraph-heavy so that later graph-RAG extraction can identify repeated relationships among diseases, organs, antibodies, complement patterns, synovial fluid categories, and treatments.

Core review principle: Rheumatology is rarely solved by a single laboratory result. The diagnosis comes from the intersection of inflammatory pattern, involved organs, time course, autoantibody profile, complement pattern, synovial fluid, imaging, and dangerous organ involvement.

I. Universal Frame for Any Rheumatologic Complaint

The first decision is not the name of the disease; it is whether the patient has an emergency. Septic arthritis, vision-threatening giant cell arteritis, pulmonary hemorrhage, rapidly progressive glomerulonephritis, cervical cord compromise, critical digital ischemia, and catastrophic antiphospholipid syndrome are treated before the full diagnostic puzzle is complete. Once emergencies are addressed, divide the presentation by joint count, inflammatory character, axial versus peripheral distribution, and extra-articular clues.

Question	High-yield interpretation	Common next step
One joint or many?	Monoarthritis suggests septic arthritis, crystal arthritis, trauma, hemarthrosis, or early inflammatory arthritis. Polyarthritis suggests RA, SLE, viral arthritis, spondyloarthritis, systemic sclerosis overlap, Sjogren, sarcoidosis, or medication-related disease.	Acutely painful swollen monoarthritis usually gets arthrocentesis before empiric anti-inflammatory closure.
Inflammatory or noninflammatory?	Inflammatory disease produces prolonged morning stiffness, swelling, warmth, nocturnal pain, fatigue, high ESR/CRP, and improvement with movement. Noninflammatory disease produces mechanical pain, brief stiffness, bony enlargement, and pain that worsens with use.	Use ESR/CRP, exam for synovitis, radiographs/ultrasound, and synovial fluid when an effusion is present.
Axial or peripheral?	Inflammatory back pain, sacroiliitis, enthesitis, dactylitis, uveitis, psoriasis, and IBD point toward spondyloarthritis. Symmetric small-joint synovitis points toward RA. Systemic symptoms with serologies may point toward SLE or vasculitis.	Ask about heel pain, uveitis, diarrhea, psoriasis, urethritis, and family history; image SI joints if axial disease is suspected.
Are organs involved?	Kidney, lung, CNS, eye, heart, skin necrosis, or digital ischemia changes urgency and treatment intensity.	Urinalysis, creatinine, CBC, chest imaging, pulse oximetry, neuro/eye evaluation, and early specialty involvement.
Could it be infection or malignancy?	Immunosuppression, fever, prosthetic joint, neutropenia, endocarditis risk, travel, HIV, hepatitis, and cancer can mimic inflammatory disease.	Culture before immunosuppression when infection is plausible. Do not start high-dose steroids casually in undifferentiated fever.

Do-not-miss rule: A hot swollen joint is septic arthritis until proven otherwise. Crystals in synovial fluid do not exclude infection, because gout or CPPD can coexist with septic arthritis.

II. Diagnostic Algorithms

Algorithm 1: Acute monoarthritis

1. Assess instability: fever, toxic appearance, severe pain with passive range of motion, prosthetic joint, immunosuppression, IV drug use, bacteremia, endocarditis signs, or inability to bear weight.
2. If an effusion is present and infection is plausible, perform arthrocentesis for cell count with differential, Gram stain, culture, and crystal analysis. Blood cultures are added when fever or bacteremia risk is present.
3. If synovial WBC is very high, neutrophil-predominant, Gram stain positive, culture positive, or clinical suspicion remains high, treat as septic arthritis with urgent drainage and antibiotics.
4. If crystals are present and cultures are negative or infection is clinically unlikely, treat acute gout or CPPD with NSAID, colchicine, or glucocorticoid depending on renal function, GI risk, anticoagulation, and infection risk.
5. If hemarthrosis is present, consider trauma, anticoagulation, inherited bleeding disorder, tumor, or neuropathic joint. If the aspirate is noninflammatory, consider osteoarthritis flare, internal derangement, or periarticular disease.

Synovial fluid category	Typical WBC pattern	Important clues	Common examples
Noninflammatory	<2,000 WBC/microliter; mostly mononuclear	Clear to straw-colored fluid; culture negative; no crystals unless coincidental.	Osteoarthritis, trauma, mechanical derangement, early avascular necrosis.
Inflammatory	2,000-50,000 WBC/microliter; neutrophils often elevated	Cloudy fluid, inflammatory pain pattern; crystals or autoimmune disease may be present.	RA, SLE arthritis, gout, CPPD, reactive arthritis, psoriatic arthritis.
Septic/highly inflammatory	Often >50,000 WBC/microliter and >75% PMNs, but lower counts occur in immunosuppression or early infection	Positive Gram stain/culture confirms; negative Gram stain does not exclude; crystals do not exclude coinfection.	Staphylococcal septic arthritis, gonococcal arthritis, prosthetic joint infection.
Crystal pattern	Inflammatory WBC range is common	Polarized microscopy: negatively birefringent needle-shaped urate for gout; positively birefringent rhomboid CPP crystals for CPPD.	Podagra, knee gout, CPPD pseudogout, elderly wrist/knee flare.

Algorithm 2: Chronic polyarthritis

1. Confirm true arthritis rather than arthralgia: look for swelling, warmth, loss of range of motion, synovial thickening, effusion, and functional limitation.
2. Define symmetry and joint distribution: RA favors symmetric MCP, PIP, wrist, and MTP involvement and usually spares DIP; psoriatic arthritis often involves DIP, dactylitis, nail disease, and asymmetry; OA involves DIP/PIP/first CMC/hips/knees/spine with mechanical features.
3. Define systemic features: malar/photosensitive rash, oral ulcers, nephritis, cytopenias, serositis, Raynaud, sicca, proximal weakness, pulmonary symptoms, uveitis, psoriasis, inflammatory bowel disease, urethritis, or constitutional symptoms.
4. Order targeted tests rather than broad panels: CBC, CMP, urinalysis with protein quantification, ESR/CRP, RF/anti-CCP for RA pattern, ANA with reflex antibodies for SLE/connective tissue disease pattern, HLA-B27 or SI imaging for axial spondyloarthritis pattern.
5. If erosive inflammatory arthritis is likely, start disease-modifying therapy early rather than treating for months with steroids alone.

Pattern	Most likely group	Clues that push the diagnosis
Symmetric MCP/PIP/wrist synovitis, morning stiffness >1 hour	Rheumatoid arthritis	Anti-CCP specificity, RF positivity, erosions, nodules, pleural disease, cervical spine instability, anemia of chronic inflammation.
Photosensitivity, oral/nasal ulcers, cytopenias, nephritis, serositis, low complement	Systemic lupus erythematosus	ANA entry clue, anti-dsDNA for renal activity, anti-Sm specificity, antiphospholipid antibodies for thrombosis/pregnancy morbidity.
Back pain before age 45, improves with exercise, sacroiliitis	Axial spondyloarthritis / ankylosing spondylitis	HLA-B27 association, uveitis, enthesitis, reduced spinal mobility, alternating buttock pain.
Psoriasis, nail pitting, dactylitis, DIP disease, enthesitis	Psoriatic arthritis	Can be asymmetric oligoarthritis or symmetric RA-like; RF usually negative; radiographs may show new bone formation.
Dry eyes, dry mouth, parotid swelling, dental caries, arthralgias	Sjogren syndrome	Anti-Ro/SSA, anti-La/SSB, RF, hypergammaglobulinemia; lymphoma risk if gland enlargement/lymphadenopathy.
Raynaud, skin thickening, reflux/dysphagia, telangiectasias	Systemic sclerosis	Limited: anticentromere and pulmonary arterial hypertension; diffuse: anti-Scl-70/RNA polymerase III, ILD, renal crisis.

Algorithm 3: Suspected vasculitis or immune organ-threatening disease

1. Recognize organ threat: hemoptysis, hypoxemia, pulmonary infiltrates, rapidly rising creatinine, active urinary sediment, mononeuritis multiplex, mesenteric ischemia, palpable purpura with systemic symptoms, retinal/visual symptoms, or digital gangrene.
2. Stabilize and exclude infection, endocarditis, embolic disease, drug reaction, malignancy, thrombotic microangiopathy, and antiphospholipid syndrome. These mimics can produce purpura, renal disease, stroke, fever, and positive inflammatory markers.
3. Map by vessel size: large vessel disease causes limb claudication, asymmetric blood pressures, bruits, headache/jaw claudication/vision loss; medium vessel disease causes nodules, livedo, renal infarcts, mesenteric ischemia, mononeuritis; small vessel disease causes purpura, glomerulonephritis, alveolar hemorrhage, sinus/lung/kidney syndromes.
4. Order targeted testing: CBC, CMP, urinalysis with microscopy and protein, ESR/CRP, ANCA with MPO/PR3, complements, hepatitis B/C, cryoglobulins, blood cultures when indicated, chest imaging, and tissue biopsy when feasible.
5. Treat organ-threatening disease urgently with high-dose glucocorticoids plus disease-specific induction therapy after reasonable infection evaluation; do not wait for every confirmatory test if vision, kidney, lung, or CNS function is at immediate risk.

III. Autoantibody and Complement Review

Autoantibodies are best interpreted as probability modifiers. A positive ANA in a patient with nonspecific fatigue and diffuse pain is weak evidence; a positive ANA with nephritis, photosensitivity, cytopenias, low complement, and anti-dsDNA is strong evidence. RF can be positive in RA, chronic infection, older age, Sjogren syndrome, and other autoimmune disease; anti-CCP is more specific for RA and predicts erosive disease. ANCA testing is meaningful in a compatible small-vessel vasculitis phenotype, not as a screening test for nonspecific aches.

Test	Most useful association	Diagnostic caution	Activity/prognosis clue
ANA	SLE, Sjogren, systemic sclerosis, mixed connective tissue disease, autoimmune myositis overlap	Sensitive but nonspecific; low-titer positives occur in healthy people and many autoimmune/infectious states.	ANA titer alone is usually not a disease activity marker.
Anti-dsDNA	SLE, especially lupus nephritis risk	Use in the right clinical context; assays vary.	Rising anti-dsDNA with falling complement can suggest active immune-complex SLE, especially nephritis.
Anti-Sm	Highly specific SLE marker	Low sensitivity; absence does not exclude SLE.	Specificity helps classification; not usually used alone for activity monitoring.

Test	Most useful association	Diagnostic caution	Activity/prognosis clue
RF	RA, Sjogren, chronic infections, mixed cryoglobulinemia	Neither sensitive nor specific enough to diagnose RA alone.	High titers in RA correlate with extra-articular disease and nodules.
Anti-CCP/ACPA	RA	Can precede symptoms; still interpret with inflammatory arthritis pattern.	Predicts erosive disease and persistent RA phenotype.
PR3-ANCA/c-ANCA pattern	GPA phenotype: sinus, lung nodules/cavitations, glomerulonephritis	ANCA-negative vasculitis exists; positive ANCA can occur in infection/drug states.	Titers correlate imperfectly with disease; treat patient, not number alone.
MPO-ANCA/p-ANCA pattern	MPA, renal-limited vasculitis, EGPA patterns	P-ANCA pattern is less specific unless antigen is MPO.	Renal/pulmonary disease drives urgency.
Antiphospholipid antibodies	APS: thrombosis, pregnancy morbidity, livedo, thrombocytopenia	Need persistent positivity; transient positivity occurs with infection.	Triple positivity and lupus anticoagulant increase thrombosis risk.
Anti-Ro/SSA and anti-La/SSB	Sjogren, SLE photosensitive/subacute cutaneous patterns, neonatal lupus risk	Anti-Ro can appear with ANA-negative or weak ANA phenotypes.	Pregnancy relevance: fetal congenital heart block surveillance in selected cases.
Anticentromere	Limited systemic sclerosis	Not all Raynaud is sclerosis.	Pulmonary arterial hypertension risk.
Anti-Scl-70/topoisomerase I	Diffuse systemic sclerosis	Clinical phenotype and organ screening remain essential.	Interstitial lung disease risk.
Anti-Jo-1/antisynthetase antibodies	Inflammatory myopathy with ILD, mechanic hands, arthritis, Raynaud	Myositis panels require weakness, CK, EMG/MRI/biopsy context.	ILD may dominate morbidity.

Complement pattern	Interpretation	Common settings
Low C3 and low C4	Classical pathway immune-complex activation; often systemic immune-complex disease.	Active SLE, lupus nephritis, cryoglobulinemic vasculitis, severe infection/endocarditis, immune-complex GN.
Low C4 with normal or mildly low C3	Classical pathway consumption or inherited/functional C1 inhibitor pathway clue.	Hereditary/acquired angioedema patterns; mixed cryoglobulinemia can have disproportionately low C4.
Normal complement with inflammatory disease	Does not exclude autoimmune disease; many inflammatory arthritides do not consume complement.	RA, spondyloarthritis, psoriatic arthritis, PMR/GCA, gout, CPPD, many ANCA vasculitides.
Rising complement after treatment	May suggest improving immune-complex activity when paired with clinical and urine improvement.	Lupus nephritis monitoring; always correlate with urinalysis and creatinine.

IV. Core Disease Illness Scripts

Illness scripts are compressed diagnostic stories. They should include age/sex tendencies, joint pattern, extra-articular clues, decisive laboratory or imaging clues, complications, and initial treatment logic. The tables below are differential diagnosis anchors rather than exhaustive pathophysiology summaries.

Disease	Anchor presentation	Key tests/clues	Treatment logic
Rheumatoid arthritis	Chronic symmetric inflammatory polyarthritis of MCP/PIP/wrists/MTPs; morning stiffness; sparing of DIP; fatigue.	Anti-CCP, RF, elevated ESR/CRP, erosions, anemia; nodules and serositis in severe seropositive disease.	Early DMARD therapy; methotrexate is typical anchor if no contraindication; avoid long-term steroid dependence.
SLE	Multisystem autoimmune disease: photosensitive rash, oral ulcers, arthritis, cytopenias, serositis, nephritis, neuropsychiatric disease.	ANA, anti-dsDNA, anti-Sm, low C3/C4, antiphospholipid antibodies, urine protein/casts.	Hydroxychloroquine foundation unless contraindicated; steroids/immunosuppressants for organ threat; renal biopsy guides nephritis regimen.
Systemic sclerosis	Raynaud plus skin thickening, reflux/dysphagia, telangiectasias, ILD/PAH/renal crisis risk.	Anticentromere limited phenotype; anti-Scl-70 ILD risk; RNA polymerase III renal crisis risk; PFT/HRCT/echo screening.	Organ-specific management; ACE inhibitor for renal crisis; avoid high-dose steroids if possible; vasodilators for Raynaud/PAH.
Sjogren syndrome	Sicca symptoms, parotid swelling, dental caries, arthralgia, fatigue; primary or secondary.	Anti-Ro/SSA, anti-La/SSB, RF, hypergammaglobulinemia, Schirmer/ocular staining, salivary gland biopsy.	Artificial tears/saliva, secretagogues, dental care; systemic immunosuppression only for extraglandular disease; watch lymphoma red flags.
Inflammatory myopathy	Symmetric proximal weakness, difficulty rising/climbing/combing hair; myalgias may be less prominent than weakness.	High CK/aldolase, AST/ALT from muscle, myositis antibodies, EMG/MRI, biopsy; dermatomyositis rash; ILD in antisynthetase.	Glucocorticoids plus steroid-sparing immunosuppression; screen for ILD and malignancy in appropriate phenotypes.
Gout	Explosive episodic arthritis, often first MTP, midfoot, ankle, knee; tophi or nephrolithiasis in chronic disease.	Needle-shaped negatively birefringent urate crystals; serum urate may be normal during flare.	Treat flare with NSAID/colchicine/steroid; allopurinol first-line ULT when indicated; target urate <6 mg/dL.
CPPD	Acute pseudogout of knee/wrist or chronic OA-like disease; older age, chondrocalcinosis.	Rhomboid positively birefringent CPP crystals; chondrocalcinosis; assess metabolic associations in younger/recurrent disease.	NSAID/colchicine/steroid for flares; no equivalent urate-lowering cure; manage OA-like damage.
Osteoarthritis	Mechanical joint pain, brief morning stiffness, bony enlargement, worse with use; DIP/PIP/first CMC, knees, hips, spine.	Radiographs: joint-space narrowing, osteophytes, subchondral sclerosis/cysts; ESR/CRP usually normal.	Exercise, weight loss, topical NSAIDs, oral NSAIDs if appropriate, injections/surgery for selected severe disease.
Fibromyalgia	Chronic widespread pain, fatigue, unrefreshing sleep, cognitive symptoms, normal inflammatory testing.	No synovitis; normal strength despite pain; screen for mimics without endless testing.	Education, graded exercise, sleep and mood treatment, CBT-informed strategies, selected neuromodulating medications; avoid chronic opioids.

V. Vasculitis and Granulomatous Disease Review

Vasculitis is best remembered by vessel size, organ pattern, and biopsy/imaging target. The same symptom, such as neuropathy or kidney injury, means different things depending on whether the patient has asthma/eosinophilia, sinus disease/lung nodules, hepatitis B exposure, cryoglobulins, purpura, or large-vessel ischemic symptoms.

Category	Diseases	Classic clues	Major danger
Large vessel	Giant cell arteritis, Takayasu arteritis	GCA: age >50, new headache, scalp tenderness, jaw claudication, vision symptoms, PMR. Takayasu: young woman, limb claudication, bruits, pulse/BP asymmetry.	GCA can cause irreversible vision loss; Takayasu can cause critical stenosis, aneurysm, stroke, renovascular hypertension.
Medium vessel	Polyarteritis nodosa, Kawasaki disease	PAN: systemic illness, livedo, nodules, mononeuritis multiplex, abdominal pain, renal ischemia without GN; hepatitis B association.	Mesenteric ischemia, renal infarction/HTN, neuropathy.
Small vessel - ANCA	GPA, MPA, EGPA	GPA: ENT, lung nodules/cavitation, GN, PR3. MPA: pulmonary-renal syndrome, MPO. EGPA: asthma, eosinophilia, sinus disease, neuropathy.	Alveolar hemorrhage, rapidly progressive GN, neuropathy, cardiac disease in EGPA.
Small vessel - immune complex	IgA vasculitis, cryoglobulinemic vasculitis, hypocomplementemic urticarial vasculitis, lupus vasculitis	Palpable purpura, arthralgias, abdominal pain, renal disease; low complement in cryoglobulins/lupus patterns.	Glomerulonephritis, GI bleeding/ischemia, neuropathy, systemic immune-complex disease.
Variable vessel	Behcet syndrome	Recurrent oral ulcers plus genital ulcers, uveitis, skin lesions, pathergy; venous/arterial thrombosis and aneurysm possible.	Eye disease, CNS disease, pulmonary artery aneurysm, thrombosis.
Granulomatous mimic/overlap	Sarcoidosis	Noncaseating granulomatous disease with hilar adenopathy, pulmonary disease, uveitis, erythema nodosum, hypercalcemia.	Cardiac conduction disease, neurosarcoidosis, severe pulmonary disease, vision threat.

GCA emergency: New headache or jaw claudication with transient or persistent visual symptoms in an older patient should trigger immediate glucocorticoid treatment while arranging urgent diagnostic evaluation. Treatment should not wait for biopsy when vision is threatened.

VI. Integrated Case Differentials

The following cases are compact review drills. Each emphasizes presentation, discriminating clues, immediate priorities, and first-line treatment concepts.

Case	Stem	Review logic
Hot swollen knee in a patient with known gout	A 62-year-old man with gout, CKD, and fever has an acutely swollen knee. Polarized microscopy shows monosodium urate crystals.	Do not stop at crystals. Fever, CKD, and a large hot joint mean septic arthritis remains possible. Send Gram stain/culture and blood cultures if febrile, consider empiric antibiotics after aspiration if suspicion is high, and arrange drainage if septic arthritis is likely.
Young woman with rash, edema, and arthritis	A 24-year-old woman has photosensitive rash, oral ulcers, symmetric arthralgias, edema, proteinuria, RBC casts, low C3/C4, positive ANA, and anti-dsDNA.	The arthritis is not the main threat; lupus nephritis is. Quantify proteinuria, assess creatinine and urine sediment, avoid nephrotoxins, involve nephrology/rheumatology, and biopsy to classify nephritis before choosing induction when feasible.
Older patient with shoulder pain and new headache	A 73-year-old has bilateral shoulder/hip girdle stiffness, high ESR/CRP, new temporal headache, and jaw fatigue while chewing.	PMR plus cranial symptoms is giant cell arteritis until proven otherwise. Treat immediately with glucocorticoids to prevent vision loss, evaluate temporal arteries and large vessels, and monitor inflammatory markers with symptoms.
Chronic back pain that improves with exercise	A 28-year-old man has low back pain for 2 years, morning stiffness, alternating buttock pain, heel enthesitis, and acute anterior uveitis.	This is an axial spondyloarthritis pattern. Ask about psoriasis and IBD, consider HLA-B27 and SI joint imaging, begin exercise/NSAID strategy if appropriate, and escalate to biologic therapy if active disease persists.
Raynaud with reflux and skin thickening	A middle-aged woman has Raynaud, puffy fingers progressing to sclerodactyly, dysphagia/reflux, telangiectasias, and dyspnea on exertion.	Systemic sclerosis requires organ screening, not only skin diagnosis. Check antibodies, PFTs/HRCT for ILD, echocardiography for pulmonary hypertension, blood pressure/creatinine/urinalysis for renal crisis risk, and treat Raynaud/reflux aggressively.
Asthma, eosinophilia, neuropathy	A patient with adult-onset asthma and chronic sinusitis develops eosinophilia, palpable purpura, foot drop, and pulmonary infiltrates.	EGPA is a key concern. Evaluate ANCA/MPO, eosinophil count, cardiac involvement, neuropathy, lung disease, and biopsy accessible tissue. Organ-threatening disease needs systemic glucocorticoids plus steroid-sparing induction strategy.
Dry eyes with parotid enlargement	A patient with long-standing sicca symptoms has recurrent parotid swelling, palpable purpura, low complement, lymphadenopathy, and weight loss.	This is not just dryness. Sjogren syndrome carries lymphoma risk; persistent gland enlargement, lymphadenopathy, purpura, low complement, cryoglobulins, and systemic symptoms need evaluation for B-cell lymphoproliferative disease.
Diffuse pain with normal exam	A patient reports widespread pain, fatigue, poor sleep, and cognitive fog. There is no synovitis, no objective weakness, normal CK, and normal ESR/CRP.	Fibromyalgia becomes likely when the exam lacks inflammatory, neurologic, and myopathic findings. Avoid repeated broad autoimmune panels. Treat with education, graded activity, sleep restoration, mood/anxiety care, and selected neuromodulators when needed.

VII. Treatment and Immunosuppression Review

Treatment is chosen by disease mechanism and organ threat. Analgesics and NSAIDs may relieve symptoms but do not prevent erosions in RA, nephritis in SLE, or vision loss in GCA. Glucocorticoids are fast but toxic; steroid-sparing therapy is the long-term strategy in most chronic immune diseases. Infection risk is cumulative and depends on disease activity, dose and duration of steroids, combination immunosuppression, age, renal disease, lung disease, diabetes, and vaccination status.

Medication group	Typical rheum use	Major monitoring / safety associations
NSAIDs	OA pain, acute gout/CPPD, spondyloarthritis symptoms, mild inflammatory pain.	GI bleeding, kidney injury, hypertension, heart failure worsening; avoid or use cautiously in CKD/anticoagulation/high GI risk.
Colchicine	Acute gout flare, gout prophylaxis during urate-lowering initiation, CPPD flare/prophylaxis in selected patients.	GI upset, cytopenias/myopathy with toxicity, dose adjust renal/hepatic impairment; interactions with strong CYP3A4/P-gp inhibitors.
Glucocorticoids	Rapid control of flares; induction/bridge in organ-threatening disease; GCA/PMR, lupus flare, vasculitis induction.	Hyperglycemia, osteoporosis, infection, adrenal suppression, myopathy, mood change, cataracts/glaucoma, avascular necrosis, weight gain, hypertension.
Hydroxychloroquine	SLE foundation, mild RA, cutaneous lupus, arthralgia; generally reduces SLE flares.	Retinal toxicity risk; baseline and periodic ophthalmologic screening; dose by actual body weight and renal risk.
Methotrexate	RA anchor DMARD, psoriatic arthritis peripheral disease, steroid-sparing in some inflammatory diseases.	CBC/LFT monitoring, mucositis, hepatotoxicity, cytopenias, pneumonitis; folic acid; teratogenic; avoid significant liver disease.
Sulfasalazine/leflunomide	RA and peripheral spondyloarthritis/PsA alternatives or combinations.	CBC/LFT monitoring; leflunomide teratogenic and long half-life; sulfasalazine GI/rash/cytopenias.
Mycophenolate	Lupus nephritis, systemic sclerosis ILD, myositis/ILD patterns.	CBC monitoring, infection risk, GI effects, teratogenicity.
Cyclophosphamide	Severe organ-threatening vasculitis, severe lupus nephritis/CNS disease, severe ILD in selected cases.	Cytopenias, infection, hemorrhagic cystitis, infertility, malignancy risk; requires careful protocols.
TNF inhibitors	RA, psoriatic arthritis, ankylosing spondylitis, IBD-associated arthritis.	TB/hepatitis screening; infection risk; avoid in certain demyelinating disease and advanced heart failure.
IL-6 inhibitors/JAK inhibitors/abatacept/rituximab	Selected RA and vasculitis/connective tissue disease contexts depending on disease.	Agent-specific infection, lab, lipid, thrombosis, hepatitis/TB, hypogammaglobulinemia, and vaccination planning issues.

Steroid complication	Why it matters clinically	Prevention/monitoring anchor
Infection risk	Dose-related risk for bacterial, viral, fungal, and opportunistic infections; symptoms may be blunted.	Use lowest effective dose, vaccinate before intense immunosuppression when possible, consider PJP prophylaxis in high-risk regimens.
Hyperglycemia/diabetes	Can precipitate severe hyperglycemia and complicate infection/wound healing.	Monitor glucose, adjust diabetes therapy, taper when possible.
Osteoporosis/fracture	Long-term steroids rapidly increase bone loss and fracture risk.	Calcium/vitamin D, weight-bearing exercise, DEXA, bisphosphonate when indicated.
Adrenal suppression	Abrupt discontinuation after prolonged therapy can cause adrenal crisis or flare.	Taper appropriately; stress-dose awareness in selected patients.
Avascular necrosis	Hip/groin pain in steroid-exposed patient should not be dismissed.	MRI if suspected; minimize cumulative dose.
Myopathy	Proximal weakness can mimic inflammatory myositis but CK is often normal.	Steroid reduction if possible; physical therapy and alternative immunosuppression.
Psychiatric effects	Insomnia, mood changes, agitation, psychosis can occur.	Warn patients, dose in morning when possible, monitor high-dose courses.
Ocular effects	Cataracts and glaucoma increase with chronic exposure.	Periodic eye care, especially if also taking hydroxychloroquine.

VIII. Red Flags and Rapid Action Triggers

Red flag	Concern	Immediate actions
Vision loss, diplopia, amaurosis fugax, jaw claudication, new temporal headache in older adult	Giant cell arteritis with ischemic optic neuropathy risk.	Start high-dose glucocorticoids immediately; urgent ophthalmology/rheumatology; temporal artery ultrasound/biopsy or imaging.
Hemoptysis, hypoxemia, diffuse infiltrates, falling hemoglobin	Pulmonary hemorrhage from ANCA vasculitis, SLE, anti-GBM, APS, infection, or DAH mimic.	Hospitalize; pulse oximetry/CT; CBC; urinalysis; ANCA/anti-GBM/complement; bronchoscopy when needed; treat infection differential.
Rising creatinine, RBC casts, dysmorphic RBCs, proteinuria	Glomerulonephritis from lupus nephritis, ANCA vasculitis, anti-GBM, cryoglobulins.	Urgent nephrology; quantify protein; serologies; renal biopsy; avoid delay in organ-threatening disease.
CNS disease: seizure, psychosis, stroke, mononeuritis multiplex	Neuro-lupus, vasculitis, APS thrombosis, infection, drug toxicity, metabolic mimic.	Stabilize; image; CSF if needed; thrombosis/infection evaluation; targeted immunosuppression only after rational exclusion of mimics.
Digital ischemia, ulcers, gangrene, severe Raynaud	Systemic sclerosis vasculopathy, APS, vasculitis, emboli, thromboangiitis, cryoglobulins.	Vascular exam, Doppler/angiography as needed, warming, vasodilators/anticoagulation or immunosuppression depending cause.
Fever on biologic, high-dose steroid, cyclophosphamide, or neutropenia	Serious infection or opportunistic infection.	Cultures/imaging; hold selected immunosuppression; empiric antimicrobials when indicated; do not assume autoimmune flare.
Severe neck pain or neurologic symptoms in RA	Atlantoaxial subluxation/cervical myelopathy.	Cervical spine imaging; avoid unsafe manipulation/intubation positioning; urgent spine/rheum evaluation.

Red flag	Concern	Immediate actions
Pregnancy morbidity or unexplained thrombosis with lupus features	Antiphospholipid syndrome.	Test lupus anticoagulant, anticardiolipin, anti-beta-2 glycoprotein I; repeat for persistence; manage thrombosis/pregnancy risk.

IX. ESR/CRP and Laboratory Interpretation Traps

ESR and CRP are inflammatory thermometers, not diagnostic labels. CRP rises and falls faster and is often more directly linked to acute inflammation or infection. ESR is influenced by age, anemia, pregnancy, kidney disease, immunoglobulins, and fibrinogen. SLE can have high ESR with relatively modest CRP unless infection, serositis, or arthritis is prominent. Normal inflammatory markers do not exclude all rheumatic disease, especially early disease, limited cutaneous disease, or fibromyalgia mimics.

Trap	Better interpretation
High ESR means autoimmune disease.	High ESR means inflammation or altered plasma proteins. Infection, malignancy, anemia, CKD, pregnancy, and monoclonal/polyclonal gammopathy can elevate ESR.
Normal ESR/CRP excludes inflammatory arthritis.	It lowers probability but does not exclude early RA, mild PsA, limited spondyloarthritis, or intermittent crystal disease. Use exam and imaging.
Positive ANA explains diffuse pain.	ANA is common and nonspecific. Diffuse pain without synovitis, cytopenias, nephritis, rash, serositis, or objective features is more often noninflammatory.
RF positive equals RA.	RF can occur in Sjogren, hepatitis C, endocarditis, older age, cryoglobulinemia, and other chronic immune stimulation. Anti-CCP plus RA pattern is stronger.
Serum urate normal excludes gout flare.	Serum urate may fall during acute flare. Crystal identification is more decisive when diagnosis is uncertain.
ANCA positive equals vasculitis.	Drug exposure, infection, inflammatory bowel disease, and lab patterns can mislead. Vasculitis diagnosis depends on compatible organ pattern and often tissue/imaging.

X. Rapid Differential Tables

Presentation	Most important differential	Discriminator
DIP arthritis	OA, psoriatic arthritis, gout/CPPD; RA is less likely	Bony nodes and mechanical pain = OA; nail pitting/dactylitis/psoriasis = PsA; abrupt inflammatory flare = crystal.
MCP/PIP/wrist synovitis	RA, SLE, viral arthritis, PsA	Anti-CCP/erosions = RA; cytopenias/nephritis/rash/low complement = SLE; acute symmetric after viral syndrome = viral.
Podagra	Gout most classic; septic arthritis still possible	Crystal analysis when uncertain or first severe flare; fever/bacteremia risk means culture.
Inflammatory back pain	Axial spondyloarthritis, IBD-associated arthritis, PsA, infection/malignancy mimic	Improves with exercise, nocturnal pain, alternating buttock pain, uveitis/enthesitis/HLA-B27/SI imaging.
Raynaud	Primary Raynaud, systemic sclerosis, SLE, MCTD, Sjogren, vasculitis, embolic/occlusive disease, drug-induced	Age of onset, ulcers, abnormal nailfold capillaries, skin thickening, autoantibodies, ischemia severity.
Proximal weakness	Inflammatory myopathy, steroid myopathy, thyroid disease, neurologic disease, electrolyte disorder	Objective weakness plus high CK/MRI/EMG = myositis; normal CK and steroid exposure suggests steroid myopathy.
Palpable purpura	Small-vessel vasculitis, IgA vasculitis, cryoglobulins, ANCA, infection, drug reaction	Urinalysis, complements, ANCA, hepatitis C, cryoglobulins, biopsy of fresh lesion.
Sicca	Sjogren, medication anticholinergic effects, dehydration, aging, hepatitis C/HIV, sarcoid, IgG4 disease	Anti-Ro/La, Schirmer/ocular staining, salivary tests, systemic features, medication review.
Serositis	SLE, RA, infection, malignancy, uremia, post-MI, autoinflammatory disease	SLE features/low complement/anti-dsDNA; RA usually severe seropositive disease; always assess infection/malignancy.

XI. Graph-RAG Concept Anchors

For graph extraction, many rheumatologic facts should connect as edges rather than isolated facts. A disease node should connect to a pattern node, organ-threat node, autoantibody node, complement node, treatment node, and medication-toxicity node. Examples: anti-dsDNA is associated with SLE and lupus nephritis activity; anti-CCP is associated with RA specificity and erosive prognosis; low C3/C4 connects to immune-complex disease; PR3-ANCA connects to GPA phenotype; high-dose glucocorticoids connect to both rapid inflammatory control and infection/osteoporosis/hyperglycemia risk.

Node A	Relationship	Node B	Reason
SLE	can cause	nephritis	Immune-complex deposition can injure glomeruli, producing proteinuria, RBC casts, and rising creatinine.
Anti-dsDNA + low complement	suggests	active immune-complex SLE	Especially when paired with urine abnormalities or systemic flare symptoms.
RA	targets	synovium	Persistent synovitis causes cartilage and bone damage, erosions, deformity, and functional loss.
Anti-CCP	predicts	erosive RA phenotype	High specificity and association with persistent/erosive disease.

Node A	Relationship	Node B	Reason
Gout	is diagnosed by	monosodium urate crystals	Needle-shaped negatively birefringent crystals in synovial fluid.
CPPD	is diagnosed by	calcium pyrophosphate crystals	Rhomboid positively birefringent crystals, often in elderly knee/wrist flares.
GCA	threatens	vision	Cranial large-vessel inflammation can cause ischemic optic neuropathy.
ANCA vasculitis	can cause	pulmonary-renal syndrome	Small-vessel capillaritis can produce alveolar hemorrhage and rapidly progressive GN.
Systemic sclerosis	can cause	renal crisis	Diffuse disease/RNA polymerase III/high-dose steroids increase risk; ACE inhibitors are central treatment.
Glucocorticoids	increase risk of	infection and osteoporosis	Risk rises with dose/duration and combination immunosuppression.

XII. Board-Style Review Prompts with Explanatory Answers

Prompt	Answer logic
A patient with symmetric MCP/PIP swelling has RF negative but anti-CCP positive disease. What is the most likely diagnosis?	Rheumatoid arthritis. RF negativity does not exclude RA; anti-CCP is more specific and supports persistent/erosive RA when the joint pattern fits.
A patient with SLE has new proteinuria and RBC casts. Which lab trend often supports active immune-complex nephritis?	Rising anti-dsDNA with falling C3/C4 supports active lupus nephritis, but renal evaluation and often biopsy guide definitive class-specific therapy.
A patient with acute knee swelling has CPP crystals. Why might antibiotics still be considered?	Crystals do not exclude septic arthritis. If fever, bacteremia risk, immunosuppression, prosthetic joint, or toxic appearance is present, culture and empiric therapy may be needed.
Which disease class is suggested by uveitis, enthesitis, inflammatory back pain, and sacroiliitis?	Spondyloarthritis. The key pattern is axial inflammation plus extra-articular features such as uveitis, psoriasis, IBD, or enthesitis.
Why are high-dose steroids dangerous in systemic sclerosis?	They can increase risk of scleroderma renal crisis, especially in diffuse disease. Steroids should be used cautiously and with clear indication.
What is the emergency in PMR symptoms plus jaw claudication?	Giant cell arteritis with risk of irreversible vision loss. Start glucocorticoids immediately while arranging diagnostic confirmation.
What differentiates fibromyalgia from inflammatory polyarthritis?	Fibromyalgia has widespread pain, fatigue, sleep disturbance, and cognitive symptoms without objective synovitis, weakness, inflammatory markers, or organ inflammation.
What does low C4 out of proportion to C3 suggest in a purpuric patient with neuropathy and hepatitis C risk?	Mixed cryoglobulinemic vasculitis is a major consideration, especially with palpable purpura, arthralgia, neuropathy, renal disease, and hepatitis C association.

Selected References and Source Notes

Uploaded rheumatology source file rheum.pdf, used as a local coverage guide for SLE, systemic sclerosis, Sjogren syndrome, mixed connective tissue disease, RA manifestations, autoantibody tables, and rheumatologic high-yield review topics.

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Aringer M, Costenbader K, Daikh D, et al. 2019 EULAR/ACR Classification Criteria for Systemic Lupus Erythematosus. Arthritis Rheumatol. 2019.

Maz M, Chung SA, Abril A, et al. 2021 ACR/Vasculitis Foundation guidelines for ANCA-associated vasculitis and for giant cell arteritis/Takayasu arteritis.

2019 ACR/Arthritis Foundation guideline for management of osteoarthritis of the hand, hip, and knee; current professional references on CPPD, fibromyalgia, Sjogren syndrome, systemic sclerosis, and inflammatory myopathy were used for synthesis.